

SHREYAS BACHIRAJU

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EDUCATION

- Carnegie Mellon University** | *Master of Science in Machine Learning* Aug. 2026 – Dec. 2027
Arizona State University | *Bachelor of Science in Informatics, GPA: 3.98* May 2026
- **Honors:** Fulton Engineering Outstanding Graduate; 2026 CRA Outstanding Undergraduate Researcher Award – Honorable Mention; 1st Place – HackHarvard 2024 (Sustainability); New American University Scholar (\$14.5k/yr); Dean’s List
 - **Leadership:** Co-Founding President of CS + Social Good at ASU (200+ members); TA – Advanced OOP; HackCMU Mentor

RESEARCH EXPERIENCE

- Carnegie Mellon University** Sep. 2026 – Present
Research Assistant, Prof. Virginia Smith
- Leading red-team research on secret loyalties in LLMs, using targeted fine-tuning to induce covert loyalty behaviors.
- Data Mining & Reinforcement Learning Lab, ASU** Sep. 2024 – Dec. 2025
Research Intern, Prof. Hua Wei
- Deployed a PyTorch–ONNX–TensorRT mixed-precision pipeline for real-time depth estimation on Jetson Nano, reducing latency by 33% and model size by 42%, while boosting throughput by 50% with 99% accuracy retained. [Poster Link](#).
 - Designed a dataset-distillation method for RL using a bi-level meta-learning pipeline in PyTorch, compressing a 100K-step PPO trajectory into 50 synthetic states and retaining 96% of expert policy performance.
 - Led an LLM-based Origin–Destination matrix optimization case study, using Chain-of-Thought prompting to iteratively update high-error traffic links while incorporating planning constraints from civil engineers, improving convergence and stability.
- ASU Center for Semiconductor Microelectronics** Jun. 2025 – Aug. 2025
Research Intern, Prof. Krishnendu Chakrabarty
- Profiled Mixtral-8×7B MoE inference on GPU–CPU systems using NVML, RAPL, and CUDA; built a thread-safe expert-swapping pipeline with explicit GPU memory management, pinned-host transfers, and CUDA stream scheduling.
 - Identified expert swapping as the primary bottleneck, contributing ~60% of energy and latency overhead; proposed an energy- and residency-aware routing policy to prefetch non-resident experts into GPU memory.

INDUSTRY EXPERIENCE

- Netradyne** May 2024 – Jul. 2024
AI/ML Intern
- Engineered an end-to-end Python and Amazon S3 pipeline to process 30,000+ dashcam videos and run depth inference in parallel, reducing total processing time by 90%.
 - Optimized monocular depth models for ADAS via batch inference and hyperparameter tuning, improving inference time by 25% and reducing Absolute Relative Error by 15%; generated 280,000+ depth maps as training data.
- Clocr Inc.** Jun. 2023 – Jul. 2023
Software Engineer Intern
- Built a RAG-based estate-planning chatbot using custom legal knowledge bases, reducing hallucinated responses by 20% through retrieval pipelines and domain-specific prompt optimization.
 - Improved chatbot helpfulness ratings by 40% in UX trials via A/B tests on prompt templates and retrieval relevance.

PUBLICATIONS

- Turnau, J., Da, L., Vo, K., Al Rafi, F., **Bachiraju, S.**, Chen, T., & Wei, H. “Joint-Local Grounded Action Transformation for Sim-to-Real Transfer in Multi-Agent Traffic Control.” *Reinforcement Learning Journal*, 6, 2271–2290, 2025. [Link](#).
- Da, L., Chen, T., Li, Z., **Bachiraju, S.**, Yao, H., Li, L., et al. “Generative AI in Transportation Planning: A Survey.” *arXiv:2503.07158*, 2025. Preprint. [Link](#).

PROJECTS

- Circle.ooo** | *Capstone Project, AI Event and Sponsorship Marketplace*
- Built a LangGraph-based multi-agent event discovery and sponsor-matching system, using web search and scraping to aggregate and rank opportunities from Eventbrite, Luma, and broader web sources, with LLM-assisted valuation and outreach.
- U-Plan – HackHarvard 2024 Winner, Sustainability** | *SAM, LLMs, GIS, Mapbox, Rasterio*
- Led a team of four to build a city-scale sustainability platform across 50+ Phoenix ZIP codes using SAM segmentation and remote-sensing analytics (NDVI, NDWI, LST) to generate LLM-driven heat-mitigation strategies.

SKILLS & COURSEWORK

- Languages:** Python, Java, C/C++, SQL, JavaScript, R, MATLAB
- ML & LLMs:** PyTorch, HuggingFace Transformers, pandas, scikit-learn, vLLM, Accelerate, LangGraph, Stable-Baselines3, OpenCV, NumPy/SciPy
- Systems / Tools:** Linux, Git, CUDA, NVML, RAPL, TensorRT, ONNX, AWS, Docker, REST
- Coursework:** Machine Learning (PhD-level), Deep Learning, NLP, Linear Algebra, Probability & Statistics, Optimization (Spring 2027), Probabilistic Graphical Models (Spring 2027), Data Structures & Algorithms, Advanced Object-Oriented Programming